# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Analytical Problem Solving

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

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**COURSE OUTCOME COVERED**
**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 20**
Question Count: 5

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**Code of Conduct**
I J O Joanna Divine (192572003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: J O Joanna Divine

**Analytical Problem Solving Rubric**

| Criteria | | Level 4 - Exemplary | Level 3 – Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Problem Interpretation | 20% | Interprets all technical requirements accurately | Interprets most requirements correctly | Partial interpretation | Misinterprets the problem |
| Analytical Reasoning | 25% | Strong reasoning with clear cause-effect analysis | Good reasoning with minor gaps | Basic analysis only | Weak or no analysis |
| Method Selection | 20% | Chooses highly appropriate virtualization and security concepts | Chooses mostly suitable concepts | Partially suitable concepts | Inappropriate approach |
| Solution Quality | 20% | Provides feasible and technically sound solution | Reasonable solution with minor issues | Basic solution with limitations | Unclear or invalid solution |
| Presentation of Analysis | 15% | Well-structured and technically precise | Generally clear | Somewhat unclear | Difficult to follow |

**Total Marks Awarded:**

# Cloud Computing Assignment

## Q1. Design a cloud data center capable of supporting disaster recovery for an online banking application.

An online banking application requires continuous availability and strict data consistency, since even a few minutes of downtime can lead to financial loss, regulatory penalties, and loss of customer trust. A well-designed cloud data center must therefore be able to withstand hardware failure, site-level disasters, and regional outages, while meeting strict Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets.

### 1. Multi-Region, Multi-Availability Zone Architecture

- Primary Data Center (Active Site): Hosts the live production workload — application servers, transaction processing engines, and the primary database cluster — spread across multiple Availability Zones (AZs) within a region so that it can survive rack- or AZ-level failures.
- Secondary / DR Data Center (Standby Site): Located in a geographically distant region to avoid correlated disasters such as floods or power-grid failures, and kept in warm or hot standby mode through continuous synchronization with the primary site.
- Global Load Balancer / DNS Failover: Continuously monitors the health of the primary region and automatically redirects customer traffic to the DR region if the primary site becomes unavailable.

### 2. Core Design Components

Compute: Auto-scaling virtual machine or container clusters (e.g., Kubernetes) are deployed identically in both regions, so the DR site can scale up quickly on failover.

Database: A distributed or clustered database uses synchronous replication within a region and asynchronous replication across regions, keeping the DR copy up to date and ready for quick promotion to primary.

Storage: Object storage with cross-region replication ensures that transaction logs, statements, and backup files exist safely in both regions.

Networking: A secure VPN or dedicated interconnect links the two sites, combined with a global load balancer for automatic traffic redirection.

Security: End-to-end encryption (in transit and at rest), hardware security module (HSM) based key management, and strict identity and access management (IAM) protect sensitive financial data across both sites.

Monitoring: Centralized logging and health-check systems detect failures early and trigger automated failover workflows without manual intervention.

### 3. Disaster Recovery Strategy

For a banking system, an Active-Active or Active-Passive (warm standby) strategy is preferred over cold DR, since the application cannot tolerate long recovery times.

- Active-Active: Both regions serve live traffic simultaneously with real-time bidirectional replication, giving the lowest possible RTO but at higher cost and complexity — suited to the most critical core-banking services.
- Active-Passive (Warm Standby): The DR region runs a smaller but fully functional replica that stays continuously updated and can be scaled up and promoted within minutes — a practical balance of cost and recovery speed for most banking workloads.

### 4. RTO/RPO and Compliance

The design should target an RTO measured in minutes and an RPO close to zero through near-synchronous replication, so that no committed transaction is lost. Automated, health-check-triggered failover removes the dependency on manual action, while regular DR drills and immutable audit logs help the bank meet regulatory business-continuity requirements.

### Conclusion

A resilient cloud data center for online banking combines multi-AZ redundancy at the primary site, a geographically separated DR site kept in near-real-time synchronization, automated global traffic routing, and strong encryption and compliance controls. This layered design allows the bank to continue operating, or to recover within minutes, even during a full regional disaster.

## Q2. Analyze the role of backup and replication in ensuring data availability.

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Data availability means that data remains accessible, accurate, and recoverable at all times, even in the presence of hardware failure, human error, corruption, or disaster. Backup and replication are the two foundational pillars of a cloud data-protection strategy. Although related, they serve different purposes and complement one another in a complete availability plan.

### 1. Role of Backup

A backup is a periodic copy of data stored separately, often in a different location or storage tier, so that data can be restored to a previous known-good state after loss, corruption, or accidental deletion.

- Protection against data loss: Guards against accidental deletion, ransomware, and application bugs that replication alone cannot fix, since replication would simply copy the corrupted data as well.
- Point-in-time recovery: Allows restoration to a specific earlier state, for example before an erroneous transaction batch was processed — important for banking audit requirements.
- Types: Full, incremental, and differential backups balance storage cost against recovery speed, with automated scheduled backups supporting compliance and long-term retention needs.
- Off-site or cross-region storage: Keeping backups in a separate region or cold-storage tier protects against disasters that affect the entire primary data center.

### 2. Role of Replication

Replication continuously copies data, either synchronously or asynchronously, from a primary system to one or more secondary systems, keeping them nearly identical in near real time.

- High availability and failover: If the primary node or region fails, a replica can be promoted almost instantly, minimizing downtime and helping meet low RTO targets.
- Load distribution: Read replicas can serve read-heavy traffic such as balance inquiries, improving performance while keeping the primary free for writes.
- Synchronous replication: Confirms a write on both sites before acknowledging the transaction, guaranteeing zero data loss (RPO close to zero) — suited to critical banking ledgers, though it adds some latency.
- Asynchronous replication: Offers better performance across long distances, with a small possible data lag, and is suitable for DR sites that are geographically distant.

## 3. Backup and Replication Compared

| Aspect | Backup | Replication |
|---|---|---|
| Purpose | Recovery from data loss or corruption | Continuous availability and fast failover |
| Frequency | Periodic (scheduled) | Continuous / near real-time |
| Protects against | Deletion, corruption, ransomware | Hardware or site failure, outages |
| Recovery speed | Slower (restore process) | Fast (near-instant failover) |
| Storage state | Historical point-in-time copies | Live, current mirror of data |

## 4. Combined Impact on Data Availability

- Replication keeps the system up and responsive during infrastructure failures by providing a ready standby copy.
- Backup allows the system to be restored to a correct state when the problem is logical or data corruption rather than infrastructure failure — something replication alone cannot undo.
- Together they help meet both RTO (through replication-driven failover) and RPO (through frequent backups and near-synchronous replication).
- For an online banking application, this combination keeps services continuously available while also protecting against silent data corruption, insider errors, or ransomware attacks.

## Conclusion

Backup and replication together form the backbone of a data availability strategy in the cloud. Replication delivers speed and continuity, keeping services online with minimal interruption, while backup delivers resilience and recoverability against corruption and human error. A robust availability strategy for a critical system such as online banking must use both in a layered, complementary manner rather than relying on either alone.